

A Spike Paper on Decompositions

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1 Introduction

The study of structural graph parameters has a long history, and the notion of tree decompositions in particular occupies a central place in it. A graph that admits a decomposition whose bags are uniformly small behaves, for many algorithmic purposes, almost like a tree, and a great number of problems that are intractable in general become solvable in linear time on such inputs. We recall the basic notions here, fix the notation used throughout, and state the results that the remainder of this note develops in detail.

Throughout, all graphs are finite, simple and undirected. For a graph G we write $V(G)$ for its vertex set and $E(G)$ for its edge set, and we abbreviate $|V(G)|$ by n whenever the graph is clear from the context. A separation of G is a pair of vertex sets whose union is $V(G)$ and between which there is no edge outside their intersection; the order of a separation is the size of that intersection. Separations of small order are the basic building blocks of every decomposition considered below.

Definition 1 (Tree decomposition). *A tree decomposition of a graph G is a pair $(T, \mathcal{B})$ consisting of a tree T and a family $\mathcal{B} = (B_t)_{t \in V(T)}$ of subsets of $V(G)$, called the bags of the decomposition, such that every vertex and every edge is covered and the bags containing any fixed vertex form a connected subtree of T . The width of the decomposition is the largest bag size minus one.*

It is worth emphasising that none of the arguments in this section depend on the particular representation chosen for the decomposition. Any of the usual encodings, whether by a tree with labelled nodes, by a nested family of separations, or by a linear order on the vertices with a bounded window, can be translated into any other with only a constant factor loss

in the width. We fix one of them for concreteness and note in passing where the translation would matter.

We now turn to the parameter itself. The minimum width of a tree decomposition of a graph is one of the most thoroughly studied quantities in structural graph theory, and it is the quantity that the whole of this note is concerned with.

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Theorem 1. *Every graph that admits a decomposition of width at most k also admits one of width at most k in which the underlying tree has at most n nodes, and such a decomposition can be computed in linear time from the given one.*

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4 Closing remarks

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carries on afterwards without interruption.

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